A residential microgrid Operational Planning & Sizing

DENSYS/ULiege | Assignment 2

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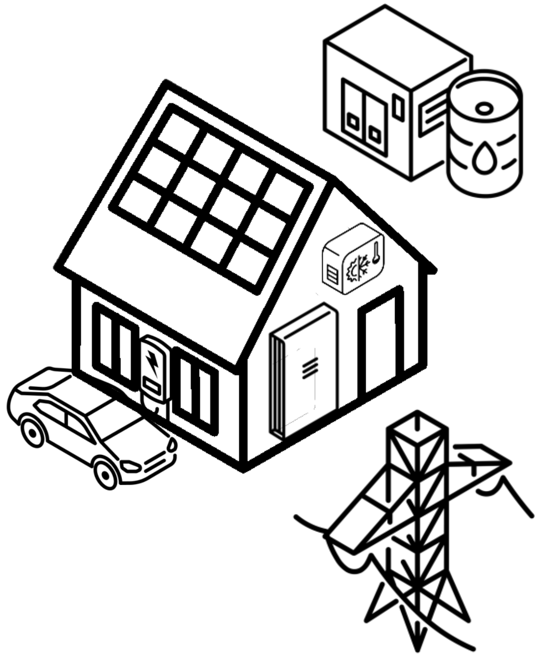
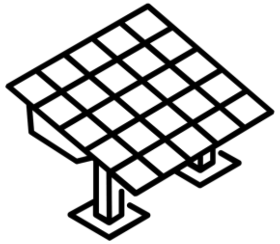
I. Case study

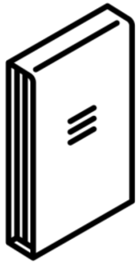
Introduction to the physical problem

Representation of the house

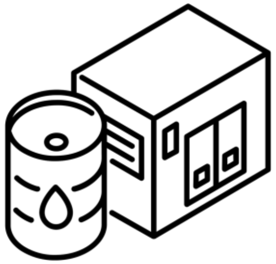




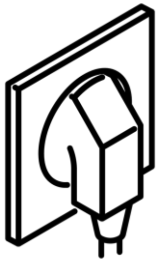
Photovoltaic panels are the main energy source



Battery storage system can be used as a slack

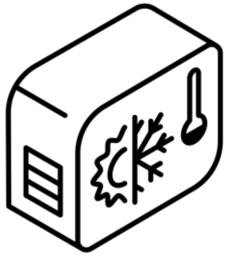


A diesel genset can also provide energy if needed

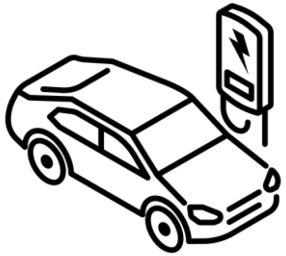


Load is fixed and should be always satisfied

The system can exchange with the grid at a given price

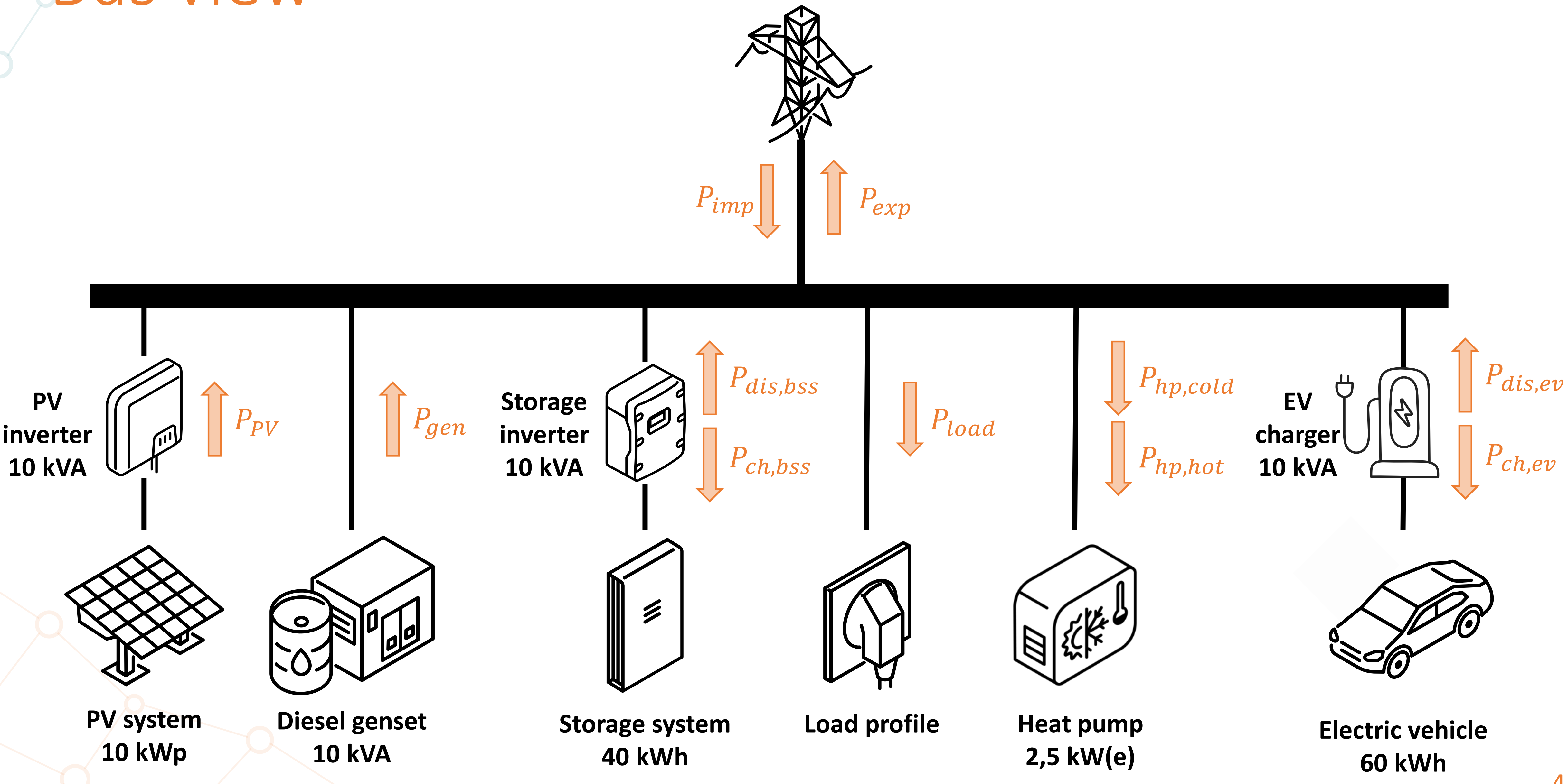


A heat pump is installed for thermal comfort



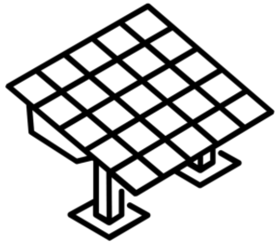
Electric vehicle connects, then should be charged

Bus view

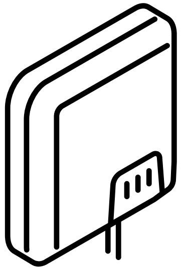


P_{imp}
P_{exp}
PV inverter 10 kVA
P_{PV}
P_{gen}
Storage inverter 10 kVA
$P_{dis,bss}$
$P_{ch,bss}$
P_{load}
$P_{hp,cold}$
$P_{hp,hot}$
EV charger 10 kVA
$P_{dis,ev}$
$P_{ch,ev}$
PV system 10 kWp
Diesel genset 10 kVA
Storage system 40 kWh
Load profile
Heat pump 2,5 kW(e)
Electric vehicle 60 kWh

Power generation



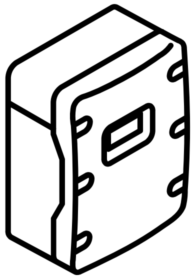
- Main source of energy
- Power profile for MPPs
  - $P_{pv,max}$
- Connected via inverter
- “Free energy”



- Inverter connected to the PV panels
- Rated capacity defines the maximum power output
  - $P_{pv,nom}$

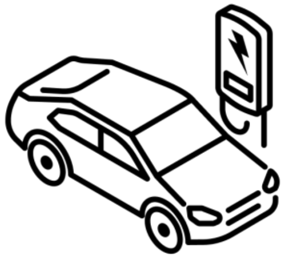
Energy storage

- Should mitigate imbalance
- Fixed capacity
  - C_{bss}
- Constant charging efficiency
  - eff_{bss}
- Bounded SOCs
  - $[SOC_{min\ bss},\ SOC_{max\ bss}]$

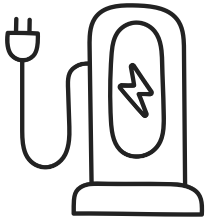


- Inverter connected to the batteries
- Limits the power output
  - $P_{bss\ nom}$

Electric vehicle

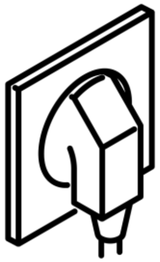


- Not always connected
  - $EV_{connection}$ (binary parameter)
- EV usage is provided with 3 vectors
  - $SOC_{ev,i}$ $\mathrm{t_{arr}}$, $\mathrm{t_{dep}}$
- Should be charged when leaving
  - $SOC_{ev\ target}$
- Fixed capacity and efficiency
  - C_{ev}, eff_{ev}
- Bounded SOCs
  - $[SOC_{min\ ev},\ SOC_{max\ ev}]$

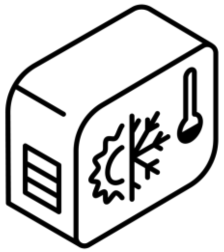


- Charger connected to the electric vehicle
- Limited charging power
  - $P_{ev\ nom}$

Fixed load and Heat pump

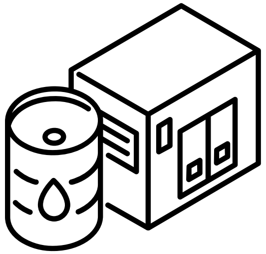


- Represents plugged-in appliances
- Defined by a profile
  - P_{load}
- Scaled by your yearly consumption
- Should always be satisfied



- Should manage temperature
- Power limited by device
  - $P_{hp\ max}$
- Home thermal capacity [kWh/°C]
  - C_{hp}
- Coefficient of performance
  - COP_{hp}
- Bounded temperature
  - $[T_{set} - \Delta_{T\ max}, T_{set} + \Delta_{T\ max}]$

Controllable generation



- Contingency usage
- Maximum/minimum power
  - $P_{max,gen}$
- Fuel costs when used
- Avoid repetitive on-off switches
- Power output should maximize efficiency

Thermal model of the house

Thermal power is equal to the electrical energy multiplied by the COP

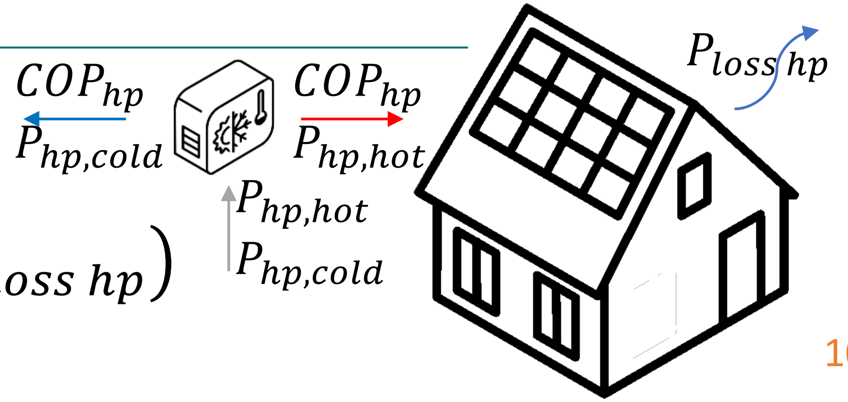
The thermal losses are subtracted to compute the net power inlet

The power is converted in energy using the time resolution

The thermal coefficient of the house gives the temperature differential

Thermal evolution of the house

$$\rightarrow T_{hp}^{t+1} = T_{hp}^{t} + \frac{\Delta_t}{C_{hp}}\left(COP_{hp}(P_{hp,hot} - P_{hp,cold}) - P_{loss\ hp}\right)$$



Objectives of the controller

This is what a good controller should achieve, in decreasing order of priority

Ensure feasibility: Power/energy bounds, demand satisfaction,…

Approach optimality: final cost, asset management, genset operation,…

Improve reliability: robustness, more realistic model of appliances,…

II. Operational Planning

Description of the first part of the homework

Objective

Using perfect forecast and given equipment sizes, define the complete optimal power profile of all assets at once.

➔ In practice, this step would be done daily, this is called day-ahead operational planning.

➔ In this part, we will consider all available data at once to get yearly costs directly.

Code diagram

Parameters

```
delta_t,SOC_i_bss, SOC_min_bss, SOC_max_bss, eff_bss
P_nom_ev, eff_ev, SOC_min_ev, SOC_max_ev, SOC_target_ev
C_ev, PI_gen, PI_imp, PI_exp, …
```

create_model()

Model object

- Contains all physical constraints of the microgrid
- Minimizes costs as objective function
- Takes decision for power set points

solve_model()

Result object

- P_load, P_pv_max (MPP ratio)
- EV profile: t_arr, t_dep, SOC_ev_i
- HP "demand": P_loss, T_set
- Asset sizes

save_results()

- Power setpoints and states:
  - P_pv P_hp
  - P_bss SOC_bss
  - P_ev SOC_ev
  - P_imp T_hp
  - P_exp
  - P_gen

check_res()
print_res()
plot_res()

Utils function

- Post processes the result object to ensure simulation results are:
  1. Feasible
  2. Optimal
  3. Looking nice and smooth

How to access data in the Results object

Inputs

- res.P_load[t] = Load power at time t
- res.P_pv_max[t] = Max available PV power at time t
- res.P_loss[t] = Power lost through house envelope
- res.T_set [t] = House temperature setpoint
- res.EV_connected[t] = EV connected at time t
- res.t_arr[t] = 1 if EV connects at time t
- res.t_dep[t] = 1 if EV leaves at time t
- res.SOC_i_ev[t] = Arrival SOC of the EV
- res.T_set[t] = Temperature setpoint for HP
- res.P_loss = Power losses through home envelope

Outputs

- res.P_pv [t] : Power of the PV system [kW]
- res.P_gen [t] : Power of the generator [kW]
- res.P_bss [t] : Power of the battery [kW]
- res.P_ev[t]: Power of the EV [kW]
- res.P_imp [t] : Power of the battery [kW]
- res.P_exp[t]: Power of the EV [kW]
- res.P_hp_hot[t]: Power for heating with HP [kW]
- res.P_hp_cold[t]: Power for cooling with HP [kW]
- res.T_hp[t]: House temperature [°C]
- res.SOC_bss[t]: Energy in the battery [kWh]
- res.SOC_ev[t]: Energy in the EV [kW]
- res. objective: Value of the objective function

Differences with last assignment

- Grid connection
- Time-step is now 15 minutes
- You handle time-series up to one year in one solve
- No power minimum for the generator
- EV can connect and disconnect multiple times
- Each DC asset is connected through an inverter with nominal power

Pay attention to solving time

- The decision horizon will go until 1 year, so the running time will grow drastically

- You should keep a linear formulation for the solution to be tractable

- Avoid non-convexity at all costs:
  - $\cancel{P_{charge} P_{discharge} = 0}$

- Use of binary variables is also unwanted if you can avoid them

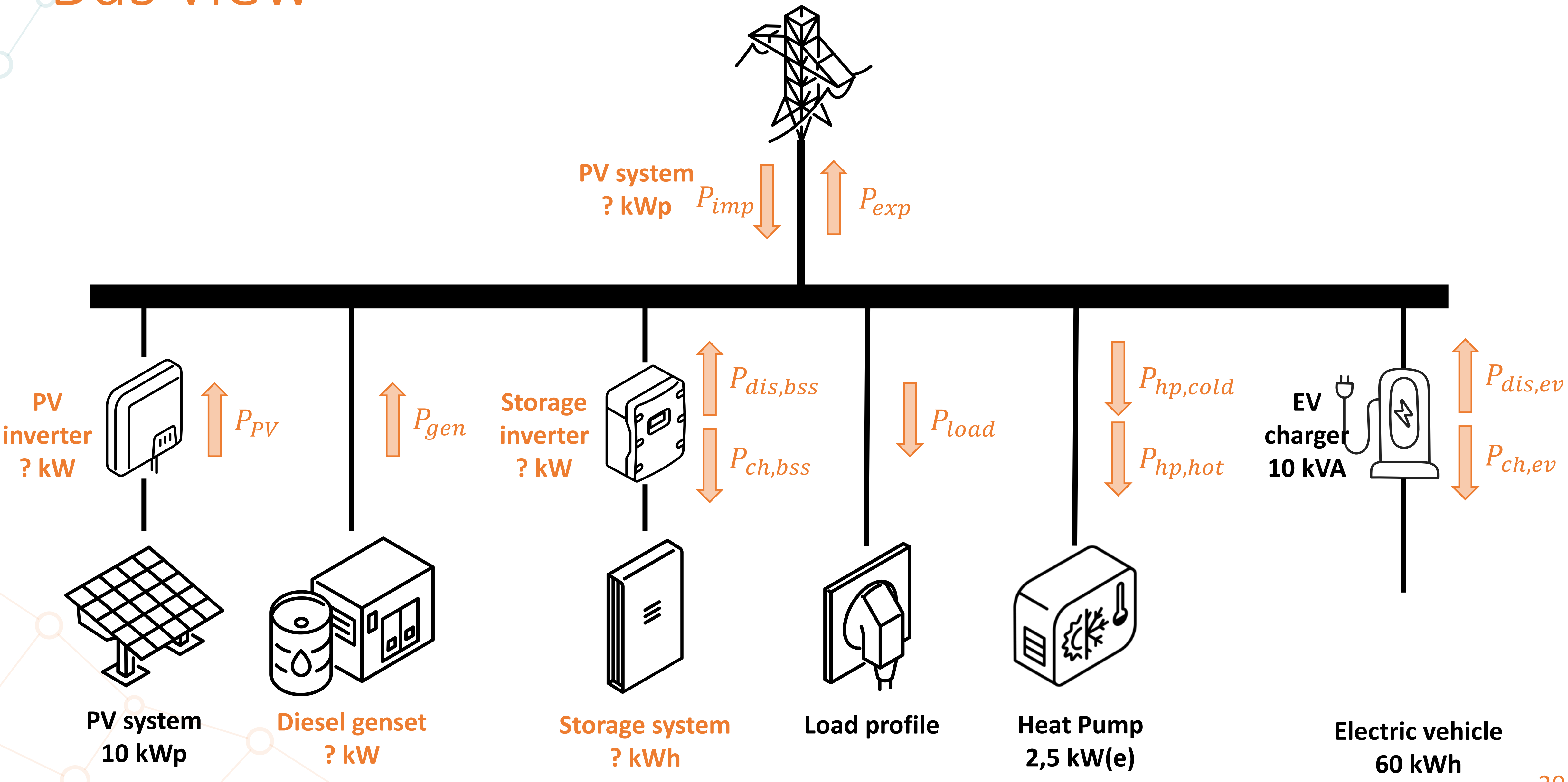
Guidelines for operational planning

0. Define the yearly load, EV_usage, and comfort temperature range
1. Solve the operational planning problem using a linear programming formulation. Optimize the use of each device to minimize OPEX.
2. Compute the yearly system costs and present some results.
3. Show how evolving energy prices would affect the results with this given system. Decorrelate import and export prices.

III. Sizing

Description of the second part of the homework

Bus view



PV system
? kWp
P_{imp}
P_{exp}
PV inverter
? kW
P_{PV}
P_{gen}
Storage inverter
? kW
$P_{dis,bss}$
$P_{ch,bss}$
P_{load}
$P_{hp,cold}$
$P_{hp,hot}$
EV charger
10 kVA
$P_{dis,ev}$
$P_{ch,ev}$
PV system
10 kWp
Diesel genset
? kW
Storage system
? kWh
Load profile
Heat Pump
2,5 kW(e)
Electric vehicle
60 kWh

Main changes from the last part

- **Asset sizes are now variables** and not parameters
- Parameters of the model now include investment horizon and investment prices for each technology
- CAPEX should be included in the objective function
- If relevant, you should also add constraints and variables to the model

Guidelines for sizing – Part 1

1. Reformulate the optimization problem of the previous section as a sizing problem. Additional constraints could be required, and the objective function must be adapted to include scaled CAPEX.

2. Compute the optimal sizing of your system, this is your base case.

3. Compare this first result with an isolated microgrid and a case where exports are not paid

4. On the base case, show how a varying CAPEX budget impacts the investment

Guidelines for sizing – Part 2

5. On the base case, make the load vary (Base, HP and EV).

6. Implement the effect of varying energy costs during the investment horizon.

7. Bonus: model CO_2 emissions and consider them in your optimization.

IV. Forecasting

Bonus part

Data and guidelines to be presented on May 6th

V. Report

Report guidelines

1. **Maximum 8 pages**

2. Pay attention to the graphs and results you present

3. Be concise, focus on your decisions and results, not on the case study
   - Try to present the result in the most adequate possible way
   - Use graphs to show dependencies
   - Use tables to summarize and compare results
   - If a proposed lead is less interesting, do not present, but justify

4. Deadline: 23:59, May 24

Remarks regarding last assignment

- Avoid generative AI
- Pay attention to the graphs you present
- Take a step back and think about your results
- Present your results in a summarized manner with focus on relevant results
- Small graph or table with numbers is often clearer than a long text